Venkata Lakshmi Parimala Pasupuleti

Staff Software Engineer · Python · Distributed Systems · PropTech · Production AI · SaaS Platform · AWS · Microservices · Cross-functional Technical Leadership

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | McLean area, Virginia — Remote

PROFESSIONAL SUMMARY

Software engineer with 8 years building production SaaS systems for property managers and owners — the exact customer base AppFolio's Investor/Owner subsegment serves. At Entrata (a direct AppFolio competitor): owned **20+ production modules end-to-end** (1M+ records, 100K+ MAU, 99.5% uptime), led Docker microservices rollout (50% speedup), AWS migration (100% PHI compliance), and shipped the first production AI pipeline on **1.6M real financial records** with quality gates and output validation. I know the property owner/investor persona — their workflows, their financial reporting needs, their compliance requirements — from 8 years of production incidents. Built **Leasel** (agentic RAG compliance, 93.6% precision on 1,240 legal PDFs) and **AgentProbe** (multi-agent LLM eval framework, LangGraph, CI/CD gates). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata

Jan 2021 – Aug 2024

Senior Software Engineer — Technical Strategy · Affordable Housing · Distributed Systems · Production AI · AWS · Cross-functional Delivery

- Owned **20+ production modules end-to-end** for property managers and owners (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints — architecture, design patterns, implementation, rollout, adoption tracking — **1M+ records, 100K+ MAU, 99.5% uptime**.
- Led **three platform-level initiatives**: Docker microservices rollout (50% reliability gain), AWS S3/RDS/Glue migration (100% PHI compliance, zero incidents), and Entrata's **first production AI pipeline on 1.6M real financial records** (PyTorch, OpenAI API) with quality gates, output validation, and HITL review before customer exposure.
- Directed technical strategy for **owner-facing financial tooling** — distribution reporting, financial statement generation, owner portal workflows — and built **HUD Section 8/HCV and LIHTC (Tax Credit)** compliance modules: AMI threshold calculations, income eligibility verification, annual/interim recertifications (HUD 50059), set-aside enforcement, and mixed-income portfolio tracking; eliminated **80% of manual workload** with a deterministic rules engine and full audit trails — zero compliance violations across 100K+ units.

Entrata

Jan 2019 – Dec 2020

Software Engineer — Compliance Automation · API Integrations · Data Pipelines

- Built compliance automation processing **millions of records** — income limits, household composition, asset verification thresholds in deterministic systems with full audit trails; built **API integrations and webhooks** across property management systems; conducted 300+ peer code reviews; mentored 5+ junior engineers.

Entrata

Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management and owner-facing workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

Leasel: Agentic Compliance Pipeline for Legal Documents

PROPTECH AGENTIC RAG

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_raq

- **93.6% precision, 87.9% F1** on 1,240 complex legal PDFs — agentic LangGraph RAG pipeline: extraction, chunking, embedding, retrieval, deterministic rules engine for fully auditable high-stakes outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge and CI/CD regression gates.

AgentProbe: Multi-Agent AI Eval Framework

MULTI-AGENT LLM-AS-JUDGE

Python · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling and inter-agent state handoffs; **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with HITL feedback loops and CI/CD regression gates blocking failing runs before production; 15-endpoint FastAPI backend, React observability dashboard.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform

HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R; FastAPI, Django, Flask, REST APIs, webhooks, API integrations, RabbitMQ, event-driven distributed systems; Ruby (familiar)

AI & Systems: LangGraph, LangChain, Agentic AI, RAG Pipelines, Multi-Agent Orchestration, Tool Calling, LLM-as-Judge, LLM Evaluation, HITL Feedback Loops, Vector Stores (ChromaDB, pgvector), OpenAI API, Anthropic API, PyTorch, BERT, NLP, system design, scalable architecture, design patterns

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, Microservices, CI/CD, GitHub Actions, PostgreSQL, MySQL, HIPAA, PHI compliance, test-driven development (TDD), refactoring, full stack web application development, Agile/Scrum, Git, Linux